

**RAJIV GANDHI COLLEGE OF ENGINEERING AND  
TECHNOLOGY**

**COMPUTER SCIENCE AND ENGINEERING**

# ***ASSIGNMENT - 4***

**Prepared by:** Mohamed B Sirajuddeen

**Reg No :** 22TD0053

**Subject Name :** E-Business

**Subject Code :** CSE 63

**Year/Sem/Sec :** 43/86/B

**Date of Submission:** 21/4/2025

**Unit :** 4

**Teacher's Sign :**

# Department Vision And Mission

## Vision

Cultivate student as technocrats, ~~researching~~ and entrepreneurs in the field of computer science and engineering.

## Missions

- To impart quality education in Computer Science and Engineering by means of learning techniques and value-added courses.
- To inculcate professional excellence and research culture by encouraging projects in cutting edge technologies through industry interactions.
- To build leadership skills, ethical values and teamwork among the students
- To strengthen the collaboration of department and industry through internships, mentorships and professional body activities.



## 1) MIME:

- \* MIME stands for multipurpose internet mail extension.
- \* It provides internet e-mail support for message containing formatted text, sound images, video and attachments.
- \* MIME is backward-compatible with SMTP message specifications and is easier to implement.
- \* Content Types are
  - Primary Type - indicate general content of material.
  - Subtype - Indicates the specific format.
- \* Composite MIME content types.
  - 1) Message - one can send the message inside another message, labeling it message/rfc822.
  - 2) Multipart - allows more than one piece of MIME to be included in message.

### MIME Encoding.

- \* Uses many different encoding methods, depending on the file type is sending.
- \* MIME provides the ability to encapsulate different content type within the body of message.
- \* The RFC-1521 version of MIME added the ability to internet e-mail to handle binary and text data, as well as multiple body parts without conversion to ASCII.

An internet electronic mail message consists of two parts.

- The header
- The body.



The header form a collection of Field / Value pairs structured is defined according to MIME.

The Multipart / signed content type:

- \* The first body part is the body part over which the digital signature was created including is MIME headers.
- \* The second body part contains the control information necessary to verify the digital signature.

Message Integrity Check (MIC)

- \* The message Integrity Check is the name given to the quantity computed over the body part with a message digest or hash function, in support of digital signature service.

- + The two related body parts are:

- 1) One for the protected Data and
- 2) One for the control information.

When creating a multipart / signed body part, the following sequence of steps describes the processing necessary.

- + The content of the body part is prepared according to local convention.
- The content is then transformed into MIME body part in canonical MIME format.
- \* The control information necessary to verify the signature.
- \* The digitally signed body part

When receiving a multipart / signed body part, the steps describes the processing to verify signature.

- \* The first body part and the control information in the second part must be prepared for signature verification process according to value of protocol.
- \* The result is made available to the user and the MIME implementation contains processing with the verified body part.
- When creating a multipart / encrypted body part,
  - \* The contents are prepared and transferred into a MIME body part in canonical MIME format, including MIME headers.
- \* The control information necessary to decrypt the body part.
- \* The encrypted body part

### MIME BODY PARTS

MIME Specifications currently support screen body types.

- \* Text,
- \* Multipart
- \* Application
- \* Message
- \* Image
- \* Audio
- \* Video.

### Text:

- \* The text body part enable a message to contain simple message data such as ASCII and can be transported using the current 7 bit message content specified within MIME.

1) Charset = US - ASCII

2) ISO-8859-1 through ISO-8859-9.



## Multipart

This part consists of several body parts containing unrelated data. They divided into subtypes

- \* Mixed
- \* Alternative
- \* parallel
- \* Digest

Body parts:

- Message
- RFC - 822
- Partial
- External body
- Image

The current subtypes are:

- 1) MPEG
- 2) GIF
  - \* Audio
  - \* Application
  - \* Postscript

## 2) Business Requirements

\* The business Requirements for secure payment processing using payment card products over both public network and private networks.

\* Security Issues regarding electronic commerce must be viewed as non-competitive in the interests of financial institutions, merchants and cardholders.

## Features

\* Confidentiality of information.

Confidentiality is ensured by the use of message encryption.

\* Integrity of data.

Payment information integrity is ensured by the use of digital signatures.

\* Cardholder account authentication.

Cardholder account authentication is ensured by the use of digital signature and cardholder certificate.

\* Merchant Authentication.

Merchant authentication is ensured by the use of digital signature and merchant certificate.

\* Interoperability.

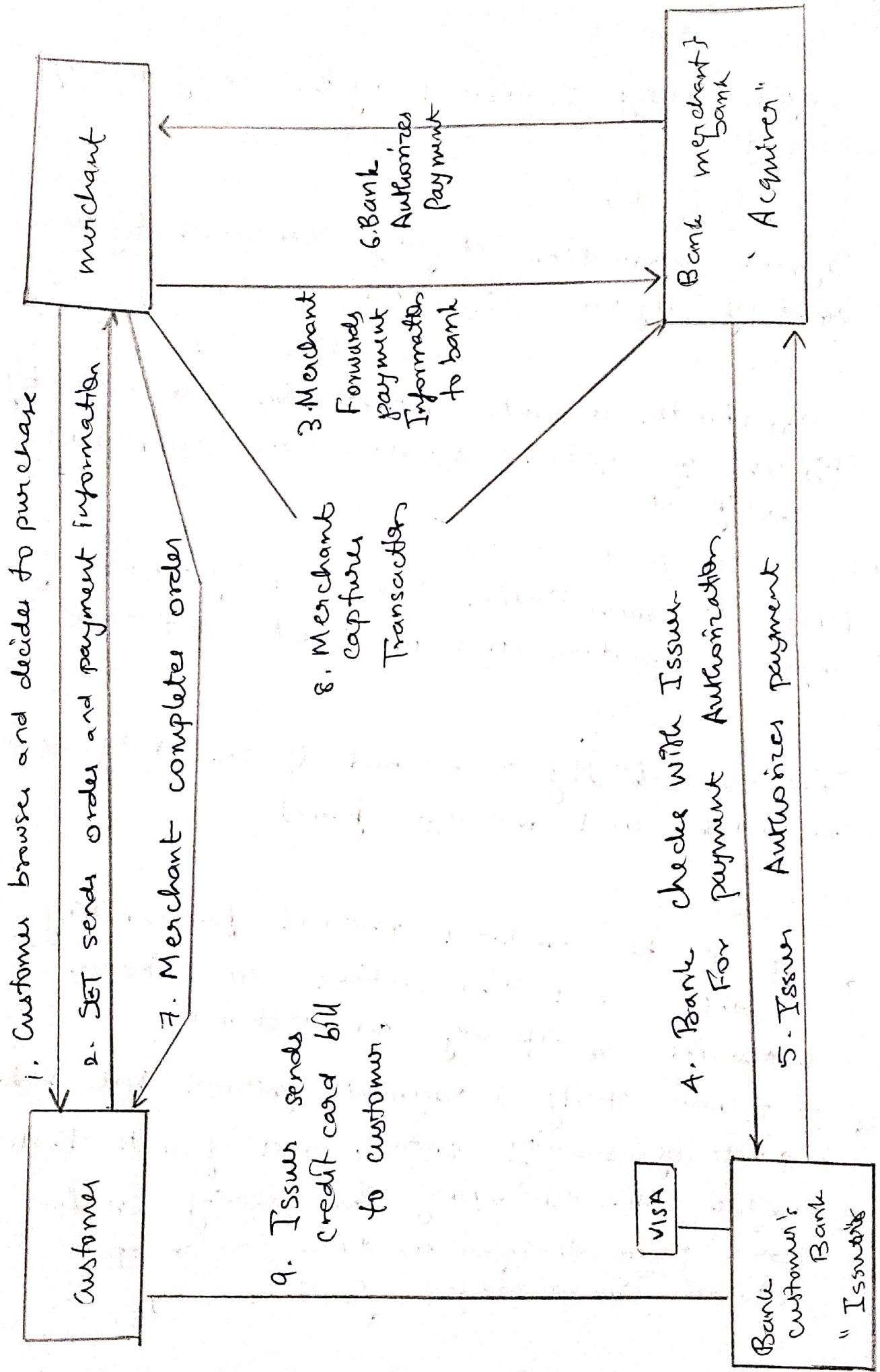
Interoperability is ensured by the use of specific protocols and message formats.

## SET Diagram:

\* It is the standard protocol for sending credit card transactions over 'home' networks specifically the merchant.

\* It is not itself a payment protocol, but rather a set of security protocol and formats that enable users to employ the existing credit card payment infrastructure on an open network in a secure fashion.







## Seven Business Requirements.

There are seven major business requirements Addressed by SET.

- 1) Provide confidentiality of payment information and enable confidentiality of order information that is transmitted along with the payment information.
- 2) Ensure integrity for all transmitted data.
- 3) Provide Authentication that a cardholder is a legitimate user of branded payment card account.
- 4) Provide Authentication that a merchant can accept branded payment card transaction through its relationship with an acquiring financial institution.
- 5) Ensure the use of best security practices and system design techniques to protect all legitimate parties of an electronic commerce transaction.
- 6) Ensure the creation of a protocol that is neither dependent on transport security mechanisms nor prevents their use.
- 7) Facilitate and encourage interoperability across software and network providers.

### 3) Payment processing:

\* This describes the flow of transaction as they are processed by various system.

\* SET defines a various of transaction protocols that utilize the cryptographic conception.

This Action describes the following Transaction

- + Cardholder registration
- + Merchant registration
- + Purchase request
- + Payment Authorization
- + Payment Capture.

### Other Transaction:

The following additional Transaction are part of these specification.

- + Certificate query
- + Purchase Inquiry
- + Purchase notification
- + Sale Transaction
- + Authorization reversal.
- + Capture reversal.
- + Credit
- + Credit reversal.

A guide to the diagrams:

#### Initial

C  
M  
P  
GA

#### Participant.

Card holder  
Merchant  
Payment Gateway  
Certificate Authority.

The following symbols are used in detailed diagram.

- + The truth of key indicate the key's owner.
- + Key with -PVII on the handle are private keys
- + Key with -PBII are public keys.
- + Keys with a diamond (◊) are signature keys.
- + Keys with small keys (k) are key-exchange keys.



- \* Dual signature, initial indicates when private key was used to create the signature certificates
- \* The initial in-scan II indicates which private key was used to sign the certificate.
- \* Symmetric key used to encrypted data.
- \* Encrypted message includes the digital envelope.

### Protocol Description

#### Certificate Authority Functions:

- \* Receive (authorities) registration requests.
- \* Process and Approve / decline requests.
- \* Issue payment certificate.

#### Optional cardholder certificate:

Payment cardholder brands and individual financial institutions will review their business needs for these functions to select a solution for implementation. Payment card brands at their option may allow cardholders to process transaction without a certificate.

#### No digital signature:

When a cardholder does not possess a signature certificate, no digital signature is generated

#### Assurance of integrity:

The recipient of data from the cardholder uses the message digest from the digital envelope to confirm the integrity of data.

Strength of cardholder certificate.

The strength depends on methods employed by payment card brand and the payment card issuer to authenticate the cardholder.

Cardholder Authentication.

If a cardholder signature certificate is not present, authentication of cardholder must be performed by means SET uses cardholder certificate to confirm.